KSP Datathon: Executing the "God-Mode" Features

Your Best Friend

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Deep Execution Guide: Building the God-Mode Stack

Here are the exact tools, actions, and AI prompts required to build the high-end post-deployment features for the KSP Datathon.

1. Execute the 3D Map (Kepler.gl)

The Tools: Install the `keplergl` and `pandas` Python libraries on your laptop.
The Action: Write a Python script that reads your 1.5 Lakh row CSV file and converts it into an interactive 3D HTML file to embed in your frontend UI.

AI Prompt: 3D Mapping

"Act as a Geospatial Data Engineer. Write a Python script using the 'keplergl' library to load a cybercrime CSV dataset. Configure the map to show 3D elevation based on transaction amounts and color heatmaps based on crime density. Save the final map as an interactive HTML file."

2. Execute the Graph Database (Neo4j)

The Tools: Create a free cloud account on **Neo4j AuraDB**. Install the `neo4j` Python driver locally.
The Action: Use a Python script to push your CSV data into the cloud, turning flat data into "Nodes" (Criminals, IP addresses) and "Edges" (Transactions) to map criminal syndicates.

AI Prompt: Graph Database

"Act as a Graph Database Architect. Write a Python script using the Python Neo4j driver to connect to a Neo4j AuraDB instance. Read a cybercrime CSV dataset and generate optimized Cypher queries to create 'Suspect' and 'Crypto_Wallet' nodes, and establish 'TRANSFERRED_TO' relationships between them."

3. Execute AI Memory (LangGraph)

The Tools: Install the `langgraph` library in your VS Code terminal.
The Action: Upgrade your existing LangChain script. LangGraph allows you to add

a "Checkpointer" (like a temporary SQLite file) that saves the conversation state so the AI acts like an ongoing agent.

AI Prompt: AI Memory

"Act as an AI Engineer. Upgrade a standard LangChain SQL Agent to use the LangGraph framework. Implement a stateful memory saver using sqlite3 as the checkpointer so the AI agent remembers previous user queries and context across different chat sessions."

4. Execute the Live Scale Test (Streaming)

The Tools: Install Python `requests` and `faker` libraries.
The Action: While your app is running on Zoho Catalyst, run a load-testing script on your laptop. It will generate fake crimes and aggressively POST them to your app's API.

AI Prompt: Live Scale Test

"Act as a QA Load Testing Engineer. Write a Python script that uses the Faker library to generate a new JSON cybercrime payload. Use the 'requests' library and threading to send a POST request with this payload to a FastAPI endpoint at a rate of 50 requests per second."

Kanglishnalli Vivarane (Execution Steps in Kanglish)

Geleya, ee ultimate features annu exact aagi hege code madodu anta illide nodi:

1. Kepler.gl (3D Map)

VS Code nalli `keplergl` install madi. AI ge prompt kottu nimma CSV data vannu interactive 3D HTML map aagi convert maduva script bareyiri. Idannu nimma website UI nalli hakabahudu.

2. Neo4j (Graph DB)

Neo4j AuraDB website ge hogi ondu free account madi. AI code balasi nimma CSV data vannu graph nodes (kallaru mattu account galu) aagi badalayisi cloud ge push madi. Idu kallara link torisutte.

3. LangGraph (AI Memory)

AI ge chat history nenapidalu `langgraph` install madi. Ee AI prompt balasi code update madi, aaga AI ge memory barutte. Obba SP nenne kelida prashne AI ge ivattu kooda nenapiruthe.

4. Live Data Streaming Test

`requests` mattu `faker` library balasi ondu hacking script tara bareyiri. Idu prati sekend ge 50 hosa fake data srushtisi nimma live Zoho API ge kaluhisutte. Idu nimma project ge entha speed ide anta KSP judges ge live aagi prove madutte.